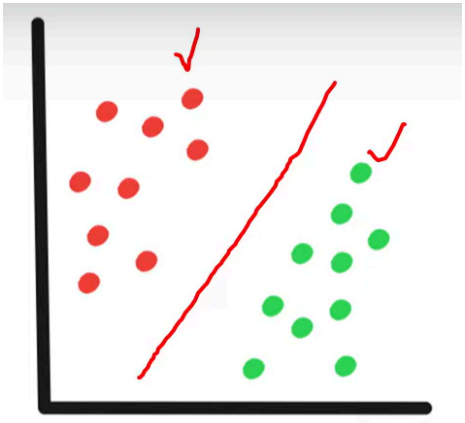
UNIT – III: Support Vector Machines – Linear & Non-linear – Metrics & Error Correction

Support Vector Machines (SVM) is a powerful supervised machine learning algorithm used for classification and regression tasks, though it is more commonly applied to classification problems. SVM works by finding a hyperplane that best divides a dataset into different classes.

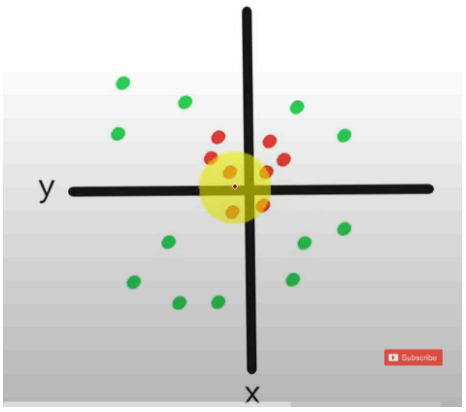
Concept of SVM

The goal of SVM is to find a decision boundary (or hyperplane) that maximizes the margin between data points of different classes. The margin is the distance between the hyperplane and the closest data points from either class, known as **support vectors**. The idea is that the wider the margin, the better the generalization ability of the model.

- **Linear SVM**: If the data is linearly separable, a straight line (in 2D), a plane (in 3D), or a hyperplane (in higher dimensions) can separate the two classes.
- **Non-linear SVM**: If the data is not linearly separable, SVM can still perform well by transforming the data into a higher-dimensional space where a linear hyperplane can be used for separation (through the **kernel trick**).



Linearly seperable



Not Linearly seperable

Margin:The margin is the distance between the decision boundary (hyperplane) and the closest data points from either class. SVM aims to maximize this margin to improve the model's generalization ability. A larger margin reduces the risk of overfitting.

Support Vectors:Support vectors are the data points that are closest to the decision boundary (hyperplane). These points are critical in defining the position and orientation of the hyperplane. They are the most important data points for constructing the model.

Hyperplane:A hyperplane is a decision boundary that separates data points of different classes. In 2D, it is a line; in 3D, a plane; and in higher dimensions, it becomes a hyperplane.

Linear SVM

When the data is linearly separable, SVM tries to find the hyperplane that best separates the two classes. The hyperplane can be represented as:

$$w \cdot x + b = 0$$

Where:

- w is the weight vector (normal to the hyperplane).
- x is the feature vector.
- b is the bias term (distance of the hyperplane from the origin).

Objective:

- Maximize the margin, which is defined as the distance from the hyperplane to the nearest data points (support vectors).
- The distance of a data point xi from the hyperplane is given by:

$$\text{Distance} = \frac{|w \cdot x_i + b|}{\|w\|}$$

Maximizing the margin is equivalent to minimizing the norm $\|w\|$, or equivalently, solving the following optimization problem:

$$\min \frac{1}{2}\|w\|^2$$

Subject to the constraints:

$$y_i(w \cdot x_i + b) \geq 1 \quad \text{for all} \quad i$$

Where y_i is the class label (+1 or -1) of the ith data point.

Non-linear SVM and Kernel Trick

In many real-world problems, the data is not linearly separable. In these cases, SVM uses the **kernel trick** to map the data into a higher-dimensional feature space where it becomes possible to find a linear hyperplane that separates the classes.

The kernel trick involves using a function $\phi(x)$ that implicitly maps the data into a higher-dimensional space without having to compute the transformation explicitly. Some commonly used kernels are:

- **Linear Kernel**: $K(x, x') = x \cdot x'$
- **Polynomial Kernel**: $K(x, x') = (x \cdot x' + c)^d$
- **Radial Basis Function (RBF) Kernel**: $K(x, x') = \exp\left(-\frac{\|x-x'\|^2}{2\sigma^2}\right)$
- **Sigmoid Kernel**: $K(x, x') = \tanh(\alpha x \cdot x' + c)$

Using the kernel trick, the SVM's decision boundary in the higher-dimensional space is a hyperplane, which is equivalent to a non-linear decision boundary in the original input space.

SVM for Multi-class Classification

SVM is inherently a binary classifier, but it can be extended to multi-class classification problems using strategies like:

- **One-vs-One (OvO)**: Create a classifier for every pair of classes. If there are kkk classes, then the total number of classifiers is k(k−1)/2
- **One-vs-Rest (OvR)**: Create a classifier for each class, where each classifier distinguishes one class from all other classes.

Metrics for SVM

To evaluate the performance of an SVM model, common metrics used are:

- **Accuracy**: Proportion of correct predictions.
- **Precision**: $\frac{TP}{TP+FP}$ – The proportion of true positives among all positive predictions.
- **Recall (Sensitivity)**: $\frac{TP}{TP+FN}$ – The proportion of actual positives correctly identified.
- **F1-Score**: Harmonic mean of precision and recall: $2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}$
- **ROC-AUC**: Measures the area under the Receiver Operating Characteristic curve, which shows the trade-off between true positive rate and false positive rate.

Error Correction and Hyperparameter Tuning

SVM models can be sensitive to hyperparameters, and improper selection can lead to poor performance. Key hyperparameters include:

- **C**: Regularization parameter that controls the trade-off between margin size and classification error.
- **Kernel**: Type of kernel to use (e.g., linear, RBF, polynomial).
- **Gamma** (for RBF kernel): A parameter that controls the influence of a single training example on the decision boundary.

Techniques like **grid search** or **random search** can be used to tune these hyperparameters and find the optimal configuration. Cross-validation is also essential to evaluate the model's generalization ability.

Advantages of SVM

- **Effective in high-dimensional spaces**: SVM works well in spaces with more features than samples.
- **Memory efficient**: SVM uses a subset of training points (support vectors) in the decision function.
- **Robust to overfitting**: Especially in high-dimensional space, given the use of the regularization parameter.

Disadvantages of SVM

- **Computationally intensive**: SVM can be slow for large datasets because of the quadratic optimization problem it involves.
- **Choice of kernel and hyperparameters**: The performance of SVM is highly dependent on the correct selection of kernel, CCC, and other hyperparameters.

Applications:

Text Classification: Used in spam detection, sentiment analysis, and document categorization.
Image Classification & Object Recognition: Applied in face recognition, handwriting recognition, and medical image analysis (e.g., tumor detection).
Biological & Biomedical: Used in gene expression data analysis, cancer detection, and protein classification.
Finance: Applied in credit scoring, fraud detection, and stock market predictions.
Speech Recognition: Used for classifying speech data in voice recognition systems.
Anomaly Detection: Detects outliers or abnormal behavior in various fields like network security and manufacturing.

Soft Margin Svm:

In real-world scenarios, the data is often not perfectly separable, which can result in overfitting if we strictly enforce a perfect separation. To handle this, SVM introduces the concept of a **soft margin**, which allows some misclassification of data points.

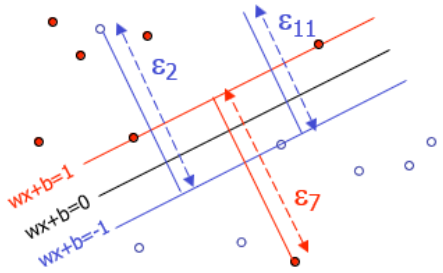
The objective function for soft margin SVM becomes:

$$\min \frac{1}{2}\|w\|^2 + C\sum_{i=1}^{n}\xi_i$$

Where:

- ξ_i is a slack variable that measures the degree of misclassification for the ith data point.
- C is a regularization parameter that controls the trade-off between maximizing the margin and minimizing the classification error. A large C gives more importance to minimizing misclassification, while a smaller C allows for a larger margin at the cost of more errors.

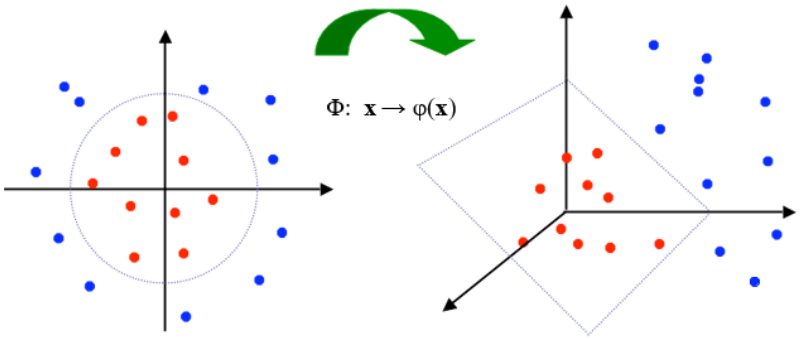
Slack variables ε_i can be added to allow misclassification of difficult or noisy examples.



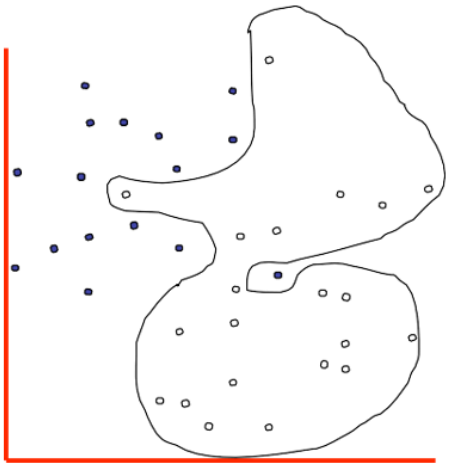
Minimize:

$$\frac{1}{2}\mathbf{w}.\mathbf{w} + C\sum_{k=1}^{R}{}''_{k}$$

Non-linear SVMs: Feature spaces



The original input space can always be mapped to some higher-dimensional feature space where the training set is separable:



Hard Margin: **requires that all data points are classified correctly**

What if the training set is noisy?

- Solution: use very powerful kernels

OVERFITTING!

Problems:

https://www.youtube.com/watch?v=YEUO81YNjSY&list=PL4gu8xQu0_5JdEqxU_FcY2x2aPte_BzF5&index=4

https://www.youtube.com/watch?v=ivPoCcYfFAw&list=PL4gu8xQu0_5JdEqxU_FcY2x2aPte_BzF5&index=5

https://www.youtube.com/watch?v=03IrkMM4E6M&list=PL4gu8xQu0_5JdEqxU_FcY2x2aPte_BzF5&index=6

https://www.youtube.com/watch?v=4-9HlWdZpJo&list=PL4gu8xQu0_5JdEqxU_FcY2x2aPte_BzF5&index=8

https://www.youtube.com/watch?v=Ll8fhDdpEbA&list=PL4gu8xQu0_5JdEqxU_FcY2x2aPte_BzF5&index=11